6 Fig. 6.1 shows a circuit containing a uniform wire P and a thermistor T that has a negative temperature coefficient.

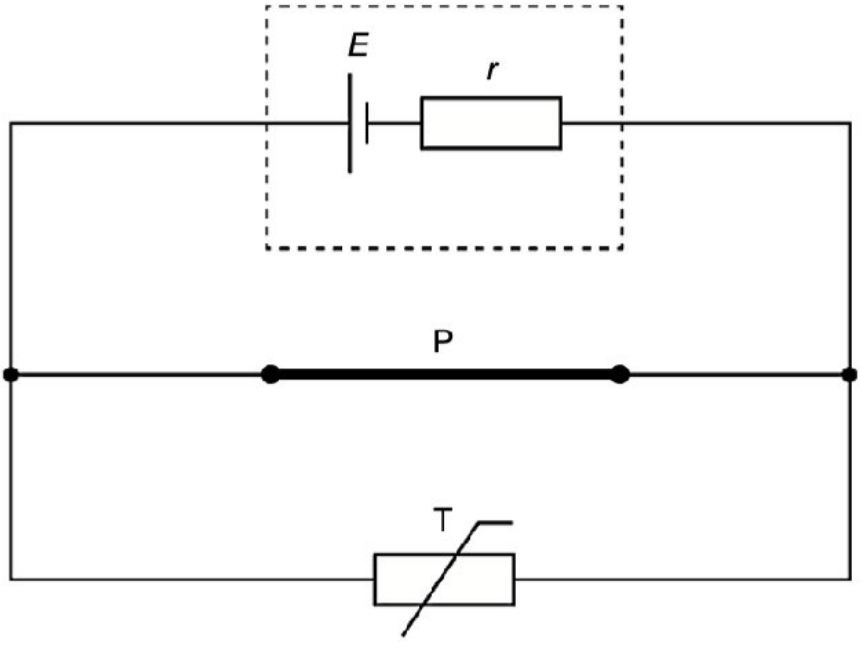


Fig. 6.1

(a) With reference to the current in the cell, explain why the current in **wire P** decreases with increasing temperature of the thermistor.

.....

.....

.....

.....

..... [3]

(b) The electromotive force (e.m.f.) E of the cell in Fig. 6.1 is 1.50 V.
The internal resistance r of the cell is 0.12 Ω.
The resistance of wire P is 6.00 Ω.
At a particular temperature of the thermistor, the current in P is 0.200 A.

Determine the resistance of the thermistor.

resistance = ... Ω [3]

(c) Wire P has a length of 0.223 m.
The **total** number of charge carriers N in the wire is 1.10×10^{20}.

Calculate the average drift velocity of the charge carriers (electrons) in the wire.

average drift velocity = ... m s^{-1} [2]

[Total: 8]